Resonant Converters: Solved Problems

EEE 339 Power Electronics · Lectures 23–24 (Hart ch. 9) · Hart Ex 9-2, 2021 Final Q6(d), 2022 Final Q3(d) (= 2024 Final Q6(c))

> **ZVS equations used (Hart 9.3, the same set printed in the 2021 paper):**
> $\omega_0 = 1/\sqrt{L_r C_r}$, $Z_0 = \sqrt{L_r/C_r}$ · $t_1 = V_s C_r / I_o$ · $t_2 = (1/\omega_0)[\sin^{-1}(V_s/(I_o Z_0)) + \pi] + t_1$ · $t_3 = (L_r I_o/V_s)\{1 - \cos[\omega_0(t_2 - t_1)]\} + t_2$ · **$V_o = V_s[1 - f_s(t_3 - t_1/2)]$** · $V_{C,peak} = V_s + I_o Z_0$
> Condition for ZVS: $I_o Z_0 > V_s$. Note that V_o **falls** as f_s rises.

Problem 1 · Hart Example 9-2 (ZVS resonant switch converter, Lecture 24)

> **Given:** $V_s = 20$ V, $L_r = 1$ µH, $C_r = 0.047$ µF, $I_o = 5$ A.
> (a) Find the switching frequency for $V_o = 10$ V. (b) Find the peak voltage across D_s when it is reverse-biased.

1. **Resonant constants.**
   $\omega_0 = 1/\sqrt{L_r C_r} = 1/\sqrt{(1\ \mu H \times 0.047\ \mu F)} =$ **4.613×10^6 rad/s**
   $Z_0 = \sqrt{L_r/C_r} = \sqrt{(1\ \mu H / 0.047\ \mu F)} =$ **4.613 Ω**
2. **Check the ZVS condition** (the capacitor voltage must swing back to zero): $I_o Z_0 = 5 \times 4.613 = 23.06$ V $> V_s = 20$ V ✓
3. **Mode 1 ($0 \to t_1$): switch opens, C_r charges linearly with I_o until $v_C = V_s$.**
   $t_1 = V_s C_r / I_o = 20 \times 0.047\ \mu F / 5 =$ **0.188 µs**
4. **Mode 2 ($t_1 \to t_2$): L_r–C_r resonance, $v_C = V_s + I_o Z_0 \sin \omega_0(t - t_1)$. t_2 is when v_C returns to zero.**
   $\sin \omega_0(t_2 - t_1) = -V_s/(I_o Z_0) = -0.8672$. The angle must be in the 3rd quadrant (after the peak), so
   $\omega_0(t_2 - t_1) = \pi + \sin^{-1}(0.8672) = \pi + 1.0495 =$ **4.1911 rad**
   $t_2 = 4.1911 / (4.613 \times 10^6) + t_1 = 0.909\ \mu s + 0.188\ \mu s =$ **1.097 µs**
5. **Mode 3 ($t_2 \to t_3$): the diode across the switch conducts, so the switch can be turned on at zero voltage.** i_L rises linearly from $I_o \cos \omega_0(t_2 - t_1)$ back to I_o:
   $t_3 = (L_r I_o/V_s)[1 - \cos \omega_0(t_2 - t_1)] + t_2 = 0.250\ \mu s \times [1 - (-0.4980)] + 1.097\ \mu s =$ **1.471 µs**
6. **Mode 4 ($t_3 \to T$): switch on, $i_L = I_o$, $v_C = 0$.** Output voltage (average of v_x): `V_o = V_s[1 - f_s(t₃ - t₁/2)]`, with $t_3 - t_1/2 = 1.377$ µs

(a) Switching frequency

$10 = 20[1 - f_s \times 1.377\ \mu s] \Rightarrow f_s = (1 - 10/20) / 1.377\ \mu s = 0.5 / 1.3771\ \mu s$

Check: $T = 1/f_s = 2.754\ \mu s > t_3 = 1.471\ \mu s$ ✓ (all four modes fit in one period).

> **(a) $f_s \approx$ 363 kHz**

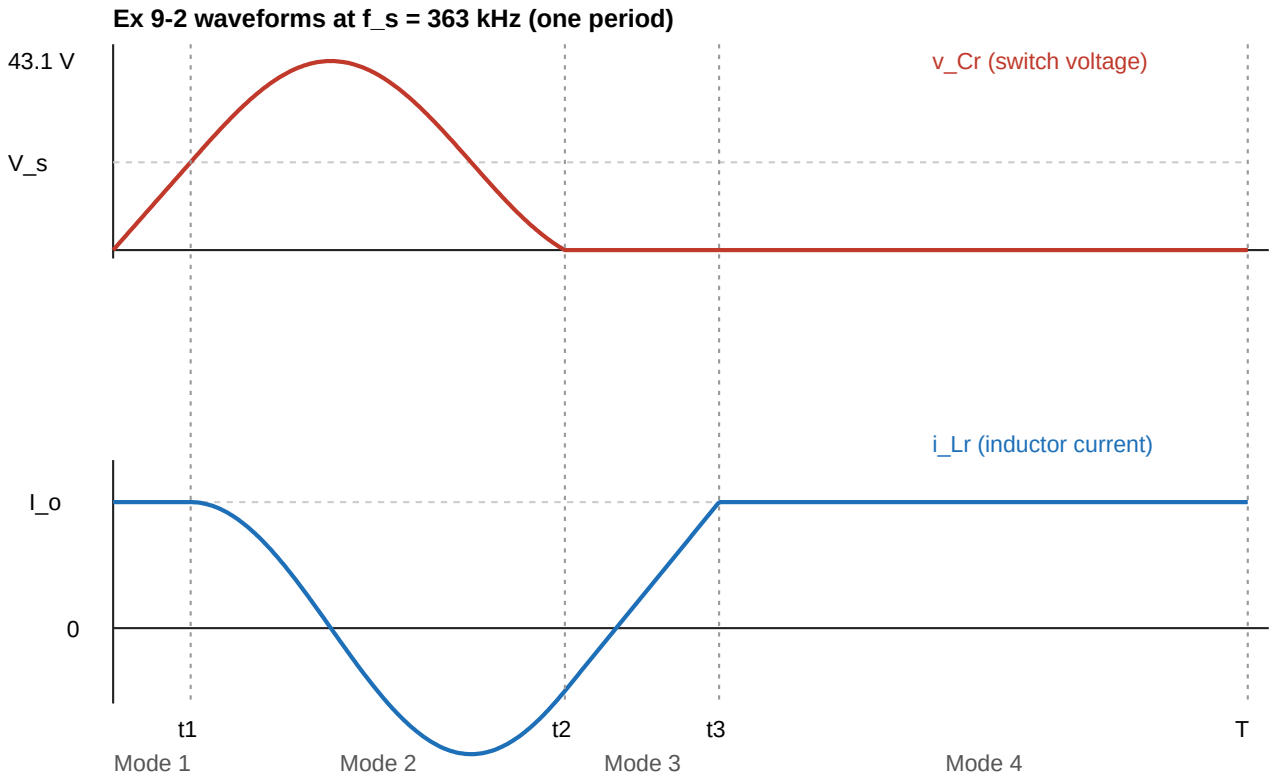
(b) Peak reverse voltage across D_s

D_s is reverse-biased while C_r is charged, so its peak reverse voltage equals the peak capacitor voltage (at $\omega_0(t - t_1) = \pi/2$ in Mode 2):

$V_{Ds,peak} = V_{C,peak} = V_s + I_o Z_0 = 20 + 5 \times 4.61 = 43.06$ V

(b) V_Ds,peak ≈ 43.1 V

Note: the slide (Hart's printed solution) writes “20 + (5)(4.61) = 33 V”. That is an arithmetic slip: 20 + 23.05 = **43.1 V**. Use 43.1 V.



Problem 2 · 2021 Final Q6(d) (ZVS resonant converter, 6 marks)

Given: V_s = 5 V, I_o = 3 A, L_r = 1 µH, C_r = 0.01 µF.
(i) Find the output voltage when f_s = 500 kHz. (ii) Find the switching frequency for V_o = 2.5 V.
The ZVS equations were printed on the question paper.

1. **Resonant constants.**
   $\omega_0 = 1/\sqrt{(L_rC_r)} = 1/\sqrt{(1\ \mu H \times 0.01\ \mu F)} =$ **1.000×10^7 rad/s**
   $Z_0 = \sqrt{(L_r/C_r)} = \sqrt{(1\ \mu H / 0.01\ \mu F)} =$ **10.000 Ω**
2. **Check the ZVS condition** (the capacitor voltage must swing back to zero): $I_oZ_0 = 3 \times 10.000 = 30.00$ V > V_s = 5 V ✓
3. **Mode 1 ($0 \to t_1$): switch opens, C_r charges linearly with I_o until v_C = V_s.**
   $t_1 = V_sC_r/I_o = 5 \times 0.01\ \mu F / 3 =$ **0.017 µs**
4. **Mode 2 ($t_1 \to t_2$): L_r-C_r resonance, $v_C = V_s + I_oZ_0 \sin \omega_0(t - t_1)$. t_2 is when v_C returns to zero.**
   $\sin \omega_0(t_2 - t_1) = -V_s/(I_oZ_0) = -0.1667$. The angle must be in the 3rd quadrant (after the peak), so
   $\omega_0(t_2 - t_1) = \pi + \sin^{-1}(0.1667) = \pi + 0.1674 =$ **3.3090 rad**
   $t_2 = 3.3090 / (1.000 \times 10^7) + t_1 = 0.331\ \mu s + 0.017\ \mu s =$ **0.348 µs**
5. **Mode 3 ($t_2 \to t_3$): the diode across the switch conducts, so the switch can be turned on at zero voltage.** i_L rises linearly from $I_o \cos \omega_0(t_2 - t_1)$ back to I_o:

$t_3 = (L_r I_o/V_s)[1 - \cos \omega_0(t_2 - t_1)] + t_2 = 0.600\ \mu s \times [1 - (-0.9860)] + 0.348\ \mu s =$ **1.539 µs**

6. **Mode 4 ($t_3 \to T$): switch on, $i_L = I_o$, $v_C = 0$.** Output voltage (average of v_x): `V_o = V_s[1 - f_s(t₃ - t₁/2)]`, with $t_3 - t_1/2 = 1.531\ \mu s$

(i) Output voltage at f_s = 500 kHz

Check: $T = 1/500\ kHz = 2.000\ \mu s > t_3 = 1.539\ \mu s$ ✓

$V_o = 5[1 - (500 \times 10^3)(1.5308 \times 10^{-6})] = 5[1 - 0.7654] = 1.173\ V$

(i) $V_o \approx 1.17$ V

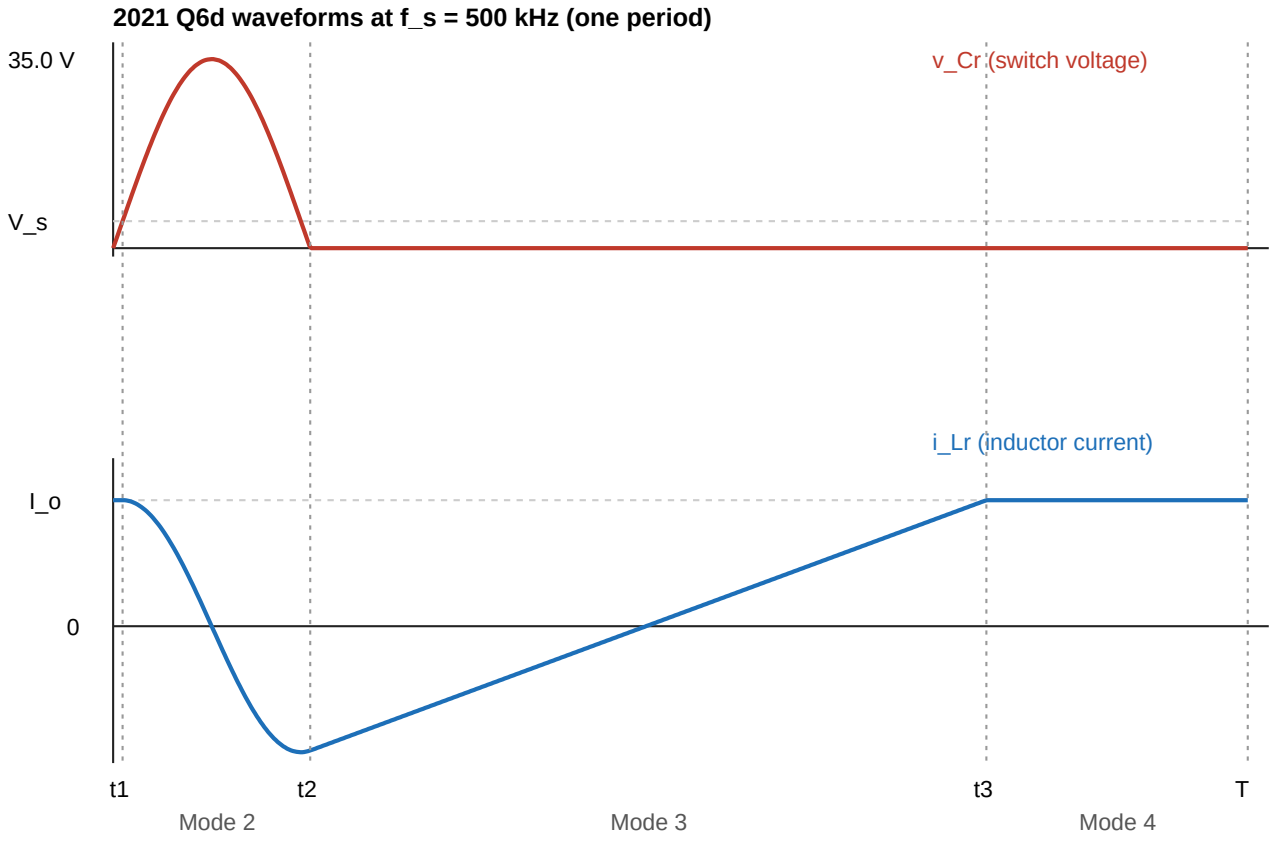
(ii) Switching frequency for V_o = 2.5 V

t_1, t_2, t_3 depend only on V_s, I_o, L_r and C_r, not on f_s, so they stay the same.

$2.5 = 5[1 - f_s \times 1.531\ \mu s] \Rightarrow f_s = 0.5 / 1.5308\ \mu s = 326.6\ kHz$

Check: $T = 3.062\ \mu s > t_3$ ✓. Lower f_s gives higher V_o, which is consistent with ZVS behaviour.

(ii) $f_s \approx 327$ kHz



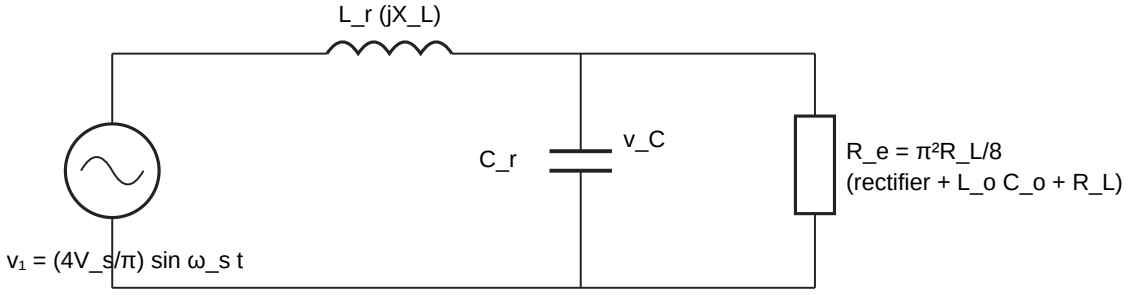
Problem 3 · 2022 Final Q3(d) (also 2024 Final Q6(c), word for word)

Given: parallel resonant DC-DC converter, $V_s = 100$ V, $L_r = 8$ µH, $C_r = 0.32$ µF, $R_L = 10$ Ω, $f_s = 120$ kHz. Output filter L_o, C_o give ripple-free output current and voltage. **Find V_o.**

This topic is **not in your slides** (it is Hart §9.6). It was asked in 2022 and 2024 with the same numbers, so it is worth learning this method.

Idea (first-harmonic method)

The switches apply a ±V_s square wave to the series L_r and parallel C_r tank. The tank filters it, so only the fundamental matters. The output rectifier, with L_o, C_o and R_L, acts as an equivalent resistor R_e in parallel with C_r. V_o is the average of the rectified capacitor voltage.



1. **Fundamental of the input square wave:** $V_1 = 4V_s/\pi = 4 \times 100/\pi =$ **127.32 V** (peak)
2. **Equivalent load resistor** (the rectifier with an L-filter draws a square-wave current): $R_e = \pi^2 R_L/8 = \pi^2 \times 10/8 =$ **12.34 Ω**
3. **Frequencies:** $\omega_0 = 1/\sqrt{(L_r C_r)} = 1/\sqrt{(8\ \mu H \times 0.32\ \mu F)} =$ **625000 rad/s** ($f_0 = 99.5$ kHz)
   $\omega_s = 2\pi \times 120$ kHz = **753982 rad/s**, so $\omega_s/\omega_0 =$ **1.206**
4. **Reactances at f_s:** $X_L = \omega_s L_r =$ **6.032 Ω**, $X_C = 1/(\omega_s C_r) =$ **4.145 Ω**
5. **Voltage transfer from source to capacitor:**
   $H = Z_p/(jX_L + Z_p)$ with $Z_p = R_e \parallel (-jX_C)$. Dividing through gives
   `H = 1 / [1 − ω_s²L_rC_r + jω_sL_r/R_e]`
   $1 - (\omega_s/\omega_0)^2 = 1 - 1.206^2 = -0.4553$; $\omega_s L_r/R_e = 6.032/12.34 = 0.4889$
   $|H| = 1/\sqrt{((-0.4553)^2 + (0.4889)^2)} = 1/\sqrt{(0.2073 + 0.2390)} =$ **1.4968**
6. **Capacitor voltage (peak):** $V_{C1} = |H| \times V_1 = 1.4968 \times 127.32 =$ **190.6 V**
7. **Output voltage** (the full-wave rectifier with the L_o filter averages $|v_C|$): $V_o = (2/\pi)V_{C1} = 0.6366 \times 190.6$

$V_o \approx$ 121 V ($V_o/V_s \approx 1.21$; the parallel resonant converter can step the voltage *up*)

One-line formula version: $V_o = (8/\pi^2)|H|V_s = 0.8106 \times 1.4968 \times 100 = 121.3$ V.

Graph check (if the paper gives Hart's V_o/V_s curves): read the curve for $Q = R_L/(\omega_0 L_r) = 2.0$ at $\omega_s/\omega_0 = 1.21$. It gives $V_o/V_s \approx 1.2$, so $V_o \approx 120$ V, which agrees.

What to write in the exam (marks checklist)

Step	Why it earns marks
Circuit sketch with L_r, C_r, switch/diode, output filter	Shows you know the topology (1 mark)
ω_0 and Z_0 calculated, ZVS/ZCS condition checked	Method marks even if the arithmetic slips later
t_1, t_2, t_3 with the quadrant explained for t_2	t_2 is where most students lose marks
$T > t_3$ check	Shows the answer is physically valid
Boxed answer with unit	Final-answer mark

All numbers were computed by script (Exam_Prep/src/resonant_solutions.py). The Problem 1 values agree with the slide solution (t_1 = 0.188 µs, t_2 = 1.10 µs, t_3 = 1.47 µs, f_s = 363 kHz).